OPTION EXERCISE VALUE

Learning Outcome

- explain the exercise value, moneyness, and time value of an option

An option buyer will exercise a call or put option at maturity only if it returns a positive payoff—that is:

- $(S_T - X) > 0$ for a call
- $(X - S_T) > 0$ for a put

If not exercised, the option expires worthless and the option buyer's loss equals the premium paid.

At any time *before* maturity ($t < T$), buyers and sellers often gauge an option's value by comparing the underlying spot price (S_t) with the exercise price (X) to determine the option's exercise value at time t. This is the option contract's value *if* the option were exercisable at time t. The exercise value for a call and a put option at time t incorporating the time value of money is the difference between the spot price (S_t) and the present value of the exercise price (PV(X)), as follows:

Call Option Exercise Value: $Max\left(0,\ S_t - X(1+r)^{-(T-t)}\right)$ 1

Put Option Exercise Value: $Max\left(0,\ X(1+r)^{-(T-t)} - S_t\right)$ 2

If we assume an exercise price, X, equal to the forward price, $F_0(T)$, the exercise value of a call option is the *same* as the value of a long forward commitment at time t ($V_t(T)$). This forward commitment was shown earlier to equal S_t − PV($F_0(T)$), *provided that* S_t > PV(X). That is, for a call option where $F_0(T) = X$:

If S_t > PV(X): S_t − PV($F_0(T)$) = Max (0, S_t − PV(X))

Note that this comparison *ignores* the upfront call option premium paid by the option buyer (c_0 at time $t = 0$).

EXAMPLE 1

Put Option Exercise Value

Consider the case of a one-year put option with an exercise price (X) of EUR 1,000 and a risk-free rate of 1%. What is the exercise value of the option in six months if the spot price (S_t) equals EUR 950?

Use Equation 2 to solve for Max (0, PV(X) − S_t):

$$X(1+r)^{-(T-t)} - S_t$$

= EUR 45.04 (= EUR 1,000$(1.01)^{-0.5}$ − EUR 950)

Discuss
Filter
Discussion
What do you want to discuss... (required)

© 2025 CFA Institute. All Rights Reserved.

Category

Pricing and Valuation of Options

Rate Your Confidence

Category

Pricing and Valuation of Options